Enterprise Student Record and Academic Management Platform

Unit-1 Mini Project

Product Engineering & Design Thinking

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Mini Project Objective

To apply Design Thinking and Product Engineering principles to design a centralized Student Record and Academic Management Platform for managing student records and academic information within a higher education institution.

1. Introduction

Educational institutions manage a large amount of academic information including student records, course registration, attendance, examinations, marks and results. When these activities are handled through disconnected or manual processes, it can become difficult to maintain accurate records, access information quickly and generate reports efficiently.

The Enterprise Student Record and Academic Management Platform is proposed as a centralized software solution for managing important academic information within a higher education institution.

The project follows a Product Engineering and Design Thinking approach. Instead of starting directly with coding, the project first focuses on understanding the educational domain, identifying stakeholders, studying their needs and problems, defining requirements, creating a product vision and planning the Minimum Viable Product (MVP).

The platform is intended to support students, faculty members, academic administrators,

examination cell staff and IT support personnel. It will provide a structured way to manage student records and academic activities while improving accessibility, usability, reliability

and security.

The main areas considered in the platform are:

- Student admission and registration
- Student record management
- Course registration
- Attendance management
- Examination management
- Result management
- Academic report generation
- Role-based access to information

The project therefore focuses on creating a clear product plan and system design that can be used for Agile planning and software development in subsequent stages.

2. Business Understanding

2.1 Academic Management Domain

A higher education institution performs several interconnected academic activities. Student information needs to be maintained throughout the academic lifecycle.

The major processes considered in this project are:

1. Student Admission

Student information is collected and recorded when a student joins the institution. This may include personal information, academic information and program details.

2. Student Registration

Students are registered in the academic system and assigned their relevant program, semester and academic information.

3. Course Registration

Students select or are assigned courses according to their academic program and semester.

4. Attendance Management

Faculty members record student attendance for different courses. Students need access to their attendance information.

5. Examination Management

The institution manages examination schedules, student participation and academic assessment.

6. Result Management

Marks and academic results are recorded and processed so that students and academic staff can access the required information.

7. Academic Report Generation

Academic information can be used to generate reports for students, faculty members, administrators and examination staff.

2.2 Problems in Academic Record Management

Manual or fragmented academic record management can create several operational problems:

- Difficulty maintaining centralized student information
- Duplicate or inconsistent records
- Time-consuming data entry
- Difficulty accessing academic information
- Errors during attendance or marks entry
- Delays in result preparation
- Difficulty generating academic reports
- Increased administrative workload
- Limited visibility of academic information
- Difficulty maintaining secure access to records

2.3 Proposed Business Solution

The proposed solution is a centralized Student Record and Academic Management Platform.

The platform will provide role-based access for different users and centralize important academic information.

The main objective is to make academic information easier to manage, access and maintain while reducing unnecessary manual effort.

3. Problem Statement

Students, faculty members and academic administrators require efficient access to accurate academic information. When student records, attendance, examinations and results are managed through separate or manual processes, users may experience delays, data-entry errors, difficulty accessing information and increased administrative workload.

Therefore, there is a need for a centralized Student Record and Academic Management Platform that can organize student information, course registration, attendance, examination and result management in a structured and accessible system.

Core Problem Statement

How might we provide students and academic staff with a centralized, secure and user-friendly platform for managing student records and academic information while reducing manual effort, errors and delays?

4. Stakeholder Analysis

Stakeholders are individuals or groups who interact with, use, manage, support or are affected by the Student Record and Academic Management Platform.

The major stakeholders identified for the proposed system are students, faculty members, academic administrators, examination cell staff, IT support personnel and college management.

4.1 Stakeholder Identification

Stakeholder	Role in the System	Main Needs
Students	Primary users of academic services	Access profile, courses, attendance, examinations and results
Faculty Members	Manage academic activities	Manage attendance, marks and student information
Academic Administrators	Manage institutional academic records	Manage students, courses and generate reports
Examination Cell	Manage examination-related activities	Manage examinations, marks and results
IT Support Personnel	Maintain the technical system	Security, maintenance, backup and technical support
College Management	Monitor institutional performance	Academic reports and overall information

4.2 Stakeholder Expectations

Students

Students expect a simple and accessible platform where they can view and manage their academic information without depending on multiple departments.

Faculty Members

Faculty members expect efficient tools for managing attendance, marks, student information and academic reports.

Academic Administrators

Administrators require centralized access to student and academic records and efficient reporting capabilities.

Examination Cell

The examination cell requires reliable tools for managing examination information, marks and result generation.

IT Support Personnel

IT support requires a secure, reliable and maintainable platform with appropriate access control and system management capabilities.

College Management

College management requires accurate academic information and reports to support monitoring and decision-making.

5. Stakeholder Matrix

Stakeholder	Influence	Interest	Importance
Students	Medium	High	High
Faculty	High	High	High
Academic Administrators	High	High	High
Examination Cell	High	High	High
IT Support	High	Medium	High
College Management	High	Medium	High

5.1 Stakeholder Engagement Strategy

Students

Collect feedback through surveys, interviews and prototype testing.

Faculty

Discuss academic workflows and requirements with faculty members.

Academic Administrators

Understand current record-management and reporting processes.

Examination Cell

Study examination, marks and result-management requirements.

IT Support

Identify security, infrastructure, maintenance and system-support requirements.

College Management

Understand business objectives, reporting requirements and expected outcomes.

6. User Persona Development

User personas represent typical users of the proposed platform. They help the project team understand user goals, responsibilities, frustrations and expectations before designing the product.

6.1 Persona 1 — Student

Name

Aarav Patil

Role

Undergraduate Student

Goals

- View academic profile
- Register for courses
- Check attendance
- View examination information
- View results

- Download academic reports

Responsibilities

- Maintain accurate personal information
- Complete course registration
- Monitor attendance
- Participate in examinations
- Check academic results

Challenges

- Academic information may be difficult to access
- Attendance information may not be immediately available
- Students may need to contact different departments for information
- Difficulty keeping track of academic records

Expectations

A simple, mobile-friendly dashboard that provides quick access to important academic information.

Important Features

- Student dashboard
- Profile management
- Course registration
- Attendance view
- Examination information
- Result view
- Report download

6.2 Persona 2 — Faculty Member

Name

Priya Sharma

Role

Assistant Professor

Goals

- View assigned students
- Manage attendance
- Enter marks
- Access course information
- Generate academic reports

Responsibilities

- Conduct academic activities
- Record attendance
- Enter assessment marks
- Monitor student performance
- Provide academic information

Challenges

- Manual attendance entry can be time-consuming
- Managing large numbers of students
- Repeated data entry
- Preparing academic reports

Expectations

A fast and reliable interface for managing students, attendance and marks.

Important Features

- Faculty dashboard
- Student list
- Attendance management
- Marks entry
- Course management
- Academic reports

6.3 Persona 3 — Academic Administrator

Name

Rahul Deshmukh

Role

Academic Administrator

Goals

- Manage student records
- Manage courses
- Monitor academic activities
- Generate reports
- Maintain accurate institutional records

Responsibilities

- Maintain student information

- **Manage academic records**
- **Coordinate academic processes**
- **Generate administrative reports**
- **Monitor system information**

Challenges

- **Large amount of student information**
- **Duplicate or inconsistent records**
- **Manual report preparation**
- **Difficulty monitoring information across departments**

Expectations

A centralized administrative dashboard providing accurate and organized academic information.

Important Features

- **Student management**
- **Faculty management**
- **Course management**
- **Academic reports**
- **Search and filtering**
- **Administrative dashboard**

8. Design Thinking

Design Thinking is a human-centered and iterative approach to solving problems. It focuses on understanding users, identifying their actual problems, generating possible solutions, creating prototypes and testing those solutions with users.

For the Student Record and Academic Management Platform, the five stages of Design Thinking are applied as follows:

Empathize → Define → Ideate → Prototype → Test

8.1 Stage 1 — Empathize

The first stage focuses on understanding the experiences, needs and problems of the people who will use the system.

The major user groups considered are:

- **Students**
- **Faculty members**
- **Academic administrators**
- **Examination cell staff**

- IT support personnel

Student Pain Points

- Difficulty accessing academic information
- Difficulty monitoring attendance
- Need to access information from different sources
- Delays in receiving academic information
- Difficulty maintaining personal academic records

Faculty Pain Points

- Time-consuming attendance management
- Managing large numbers of students
- Manual marks entry
- Difficulty preparing reports
- Repeated data entry

Administrator Pain Points

- Managing large amounts of student information
- Maintaining accurate records
- Duplicate or inconsistent information
- Time-consuming report generation
- Difficulty monitoring academic activities

Examination Cell Pain Points

- Managing examination information
- Collecting and validating marks
- Preparing results
- Maintaining examination records

8.2 Stage 2 — Define

The information collected during the Empathize stage is used to define the main problem.

Defined Problem

Students, faculty members and academic administrators need a centralized platform to manage student records and academic information because fragmented and manual processes can cause delays, errors and difficulty accessing accurate information.

User Need Statement

Students need a simple way to access their academic information, while faculty and administrators need efficient tools to manage academic records and generate reports.

8.3 Stage 3 — Ideate

During the Ideate stage, possible solutions are generated without immediately focusing on technical limitations.

The following ideas were identified:

1. **Centralized student profile management**
2. **Online student registration**
3. **Online course registration**
4. **Digital attendance management**
5. **Online examination management**
6. **Marks entry system**
7. **Automatic result generation**
8. **Academic report generation**
9. **Role-based dashboards**
10. **Academic notifications**
11. **Search and filtering of student records**
12. **Secure authentication and authorization**

Proposed Solution

The main idea is to develop a centralized platform where each user receives features according to their role.

Student → Academic information and self-service features

Faculty → Attendance, marks and student management

Administrator → Student, course and academic record management

Examination Cell → Examination, marks and result management

IT Support → Security, system maintenance and technical support

8.4 Stage 4 — Prototype

The Prototype stage converts the proposed ideas into simple representations of the product before full development.

A low-fidelity prototype will be prepared for the major system screens.

Proposed Prototype Screens

1. Login Screen

The login screen provides secure access to the platform.

Main elements:

- **User ID / Username**
- **Password**

- **Login button**
- **Forgot password option**
- **Role-based access**

2. Student Dashboard

The student dashboard provides quick access to:

- **Profile**
- **Courses**
- **Attendance**
- **Examination**
- **Results**
- **Academic reports**

3. Attendance Screen

The attendance screen displays:

- **Subject name**
- **Classes conducted**
- **Classes attended**
- **Attendance percentage**
- **Overall attendance**

4. Results Screen

The results screen displays:

- **Subject**
- **Marks**
- **Grade**
- **Semester performance**
- **CGPA**
- **Academic report download option**

5. Faculty Dashboard

The faculty dashboard provides:

- **Student list**
- **Course list**
- **Attendance management**
- **Marks entry**
- **Student performance**
- **Report generation**

8.5 Stage 5 — Test

The prototype should be tested with representative users to determine whether the

proposed interface and workflows are understandable and useful.

Testing Questions

1. **Can students easily find their attendance?**
2. **Can students locate their results without assistance?**
3. **Is the dashboard easy to understand?**
4. **Can faculty members quickly record attendance?**
5. **Can faculty members enter marks easily?**
6. **Can administrators find student records quickly?**
7. **Are the available features clearly organized?**

Prototype Testing Feedback

Feedback	Proposed Improvement
Students want attendance visible immediately	Add attendance summary to dashboard
Students need quick access to results	Add results shortcut
Faculty needs faster attendance entry	Add dedicated attendance button
Administrators need faster record searching	Add search and filtering
Users prefer simple navigation	Use role-based dashboards

8.6 Design Thinking Outcome

The Design Thinking process helped identify the major users, understand their problems, define the core problem, generate possible solutions and design the initial prototype.

The proposed platform will therefore be developed around actual user needs rather than starting directly with technical implementation.

9. IBM Design Thinking

IBM Design Thinking is an approach for applying human-centered design to large-scale software products. It focuses on understanding user outcomes, continuously improving the product and enabling collaboration between different roles.

For the Student Record and Academic Management Platform, the following three principles are applied.

9.1 Focus on User Outcomes

The platform should focus on the results users want to achieve rather than only focusing on technical features.

Example

Instead of defining the goal as:

"Create an attendance database."

The user-focused outcome is:

"Allow students to check their current attendance quickly and easily."

Similarly:

Technical Feature	Desired User Outcome
Student database	Students can access accurate personal information
Attendance module	Students can quickly check attendance
Result module	Students can view academic results easily
Report generation	Administrators can generate reports efficiently
Course registration	Students can complete registration without unnecessary manual processes

The success of the platform should therefore be measured by how effectively it solves user problems.

9.2 Restless Reinvention

Restless Reinvention means continuously improving the product based on user feedback, system usage and changing requirements.

The platform should not be considered a finished product after its first release.

The improvement cycle can include:

1. **Collect user feedback**

2. **Analyze user behaviour**
3. **Identify problems**
4. **Prioritize improvements**
5. **Develop new features**
6. **Test the changes**
7. **Release improvements**
8. **Collect feedback again**

Example

If students report that checking attendance requires too many steps, the team can redesign the dashboard and place an attendance summary directly on the home screen.

This creates continuous product improvement.

9.3 Diverse Empowered Teams

The platform should be developed by a cross-functional team where different members contribute their expertise.

Proposed Team

Team Role	Main Responsibility
Product Owner	Defines product goals and priorities
UI/UX Designer	Designs user interfaces and user experience
Frontend Developer	Develops the user interface
Backend Developer	Develops application logic and APIs
Database Developer	Designs and manages the database
QA Engineer	Tests functionality and quality
DevOps Engineer	Handles CI/CD, deployment and infrastructure

A diverse team helps identify technical, business, usability and operational requirements from different perspectives.

9.4 IBM Design Thinking Application

The three principles are applied together:

Focus on User Outcomes

↓

Understand what students, faculty and administrators need to achieve.

↓

Restless Reinvention

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Continuously improve the platform using feedback and observations.

↓

Diverse Empowered Teams

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Enable different team members to collaboratively design, develop and improve the product.

10. IBM Loop Documentation

The IBM Loop used in the project follows the continuous cycle:

Observe → Reflect → Make

The purpose of the loop is to continuously learn from users, analyze the information and create improvements.

10.1 Observe

The team observes how users interact with the academic management process.

The following can be observed:

- **How students access academic information**
- **How faculty members record attendance**
- **How marks are entered**
- **How administrators manage student records**
- **Where users experience delays**
- **What errors occur**
- **What feedback users provide**

Example Observation

Students may need to navigate through multiple pages to find their attendance information.

10.2 Reflect

The team analyzes the observations and determines what should be improved.

For the attendance example:

Observation: Students have difficulty finding attendance information.

Reflection: Attendance is an important and frequently accessed feature and should be more visible.

Decision: Place an attendance summary directly on the student dashboard.

10.3 Make

The team creates and implements the improvement.

For the example:

- **Design an attendance summary card**
- **Display overall attendance percentage**
- **Provide a link to detailed attendance**
- **Test the new design with users**

10.4 IBM Loop Example

Observe

Students have difficulty finding attendance information.

↓

Reflect

Attendance should be immediately visible after login.

↓

Make

Add an attendance summary card to the student dashboard.

↓

Observe Again

Check whether students can now find attendance more easily.

↓

Reflect Again

Analyze user feedback.

↓

Make Again

Improve the interface based on the feedback.

10.5 IBM Loop Outcome

The IBM Loop ensures that the Student Record and Academic Management Platform remains user-centered and continuously improves throughout its development lifecycle.

11. Business Requirements Document (BRD)

11.1 Document Purpose

The purpose of this Business Requirements Document is to define the business needs, objectives, scope and expected outcomes of the Enterprise Student Record and Academic Management Platform.

The document provides a foundation for designing and developing a centralized system for managing student records and academic information.

11.2 Business Objective

The main business objective is to provide a centralized platform that simplifies the management of student academic information and reduces the difficulties associated with manual or fragmented academic processes.

The platform aims to:

- **Centralize student academic records**
- **Improve accessibility of academic information**
- **Reduce manual data entry**
- **Reduce data inconsistency and errors**
- **Simplify attendance management**
- **Improve examination and result management**
- **Reduce administrative workload**
- **Provide efficient academic reporting**
- **Improve the overall user experience**

11.3 Business Scope

In Scope

The initial platform will cover:

- Student registration
- Student profile management
- Course registration
- Attendance management
- Examination management
- Marks management
- Result generation
- Academic report generation
- Role-based access
- Basic notifications
- Administrative management

Out of Scope

The following features are not part of the initial MVP:

- Online fee payment
- Hostel management
- Transportation management
- Library management
- Advanced artificial intelligence recommendations
- Complex predictive analytics
- Third-party payment integration

These features may be considered in future releases.

11.4 Business Users

The main business users are:

1. Students
2. Faculty Members
3. Academic Administrators
4. Examination Cell Staff
5. IT Support Personnel
6. College Management

11.5 Business Requirements

BR-01: Centralized Student Records

The system should provide a centralized location for maintaining student academic information.

BR-02: Student Registration

The system should support registration of students and maintenance of their academic

profiles.

BR-03: Course Registration

The system should allow students and authorized academic staff to manage course registration.

BR-04: Attendance Management

The system should provide a digital mechanism for faculty members to record attendance and students to view their attendance.

BR-05: Examination Management

The system should support management of examination-related information.

BR-06: Result Management

The system should support marks entry and result generation.

BR-07: Academic Reporting

The system should provide academic reports for authorized users.

BR-08: Role-Based Access

The system should provide different access permissions according to user roles.

BR-09: Data Security

Academic information should be protected against unauthorized access.

BR-10: Information Accessibility

Authorized users should be able to access required academic information easily and efficiently.

11.6 Business Benefits

The proposed platform is expected to provide the following benefits:

Reduced Administrative Work

Digital workflows can reduce repetitive manual record management.

Improved Accuracy

Centralized records can reduce duplicate and inconsistent information.

Faster Information Access

Students and staff can access information through the platform instead of depending on separate manual processes.

Better Academic Monitoring

Faculty and administrators can monitor attendance, marks and academic performance more effectively.

Improved Reporting

Academic reports can be generated from centralized information.

Better User Experience

A role-based interface can provide users with only the features relevant to their responsibilities.

11.7 Business Assumptions

The project assumes that:

- Students and faculty have access to the platform.
- Users have valid login credentials.
- Academic departments provide accurate information.
- Authorized staff maintain academic records.
- The institution has suitable network and computing infrastructure.
- Users receive basic training when required.

11.8 Business Constraints

Potential constraints include:

- Limited development time
- Availability of accurate academic data
- Institutional policies
- Data privacy requirements
- Infrastructure limitations
- User adoption and training
- Integration with existing systems

11.9 Business Success Criteria

The project will be considered successful when:

- Students can access their academic information easily.
- Faculty can manage attendance and marks efficiently.
- Administrators can manage student records centrally.
- Examination staff can manage examination and result information.
- Authorized users can generate required academic reports.

- Academic information is protected through appropriate access control.
- The platform provides a simple and usable user experience.

11.10 Expected Business Outcome

The expected outcome is a centralized Student Record and Academic Management Platform that improves the efficiency, accessibility and organization of academic information management within a higher education institution.

12. Functional Requirements Specification (FRS)

12.1 Introduction

The Functional Requirements Specification defines the functions and services that the Student Record and Academic Management Platform must provide to its users.

The requirements describe what the system should do to support students, faculty members, academic administrators and examination cell staff.

12.2 User Authentication and Access

FR-01: User Login

The system shall allow registered users to log in using valid credentials.

FR-02: Role-Based Access

The system shall provide different access permissions according to the user's role.

The main roles are:

- Student
- Faculty
- Academic Administrator
- Examination Cell Staff
- IT Support

FR-03: Logout

The system shall allow users to securely log out of their accounts.

12.3 Student Management

FR-04: Student Registration

The system shall allow authorized administrators to register students and create their academic records.

FR-05: Student Profile Management

The system shall allow authorized users to create, view and update student profile information.

FR-06: Student Search

The system shall allow authorized staff to search for students using relevant information such as student ID, name or program.

12.4 Course Management

FR-07: Course Management

The system shall allow authorized administrators to create, update and manage course information.

FR-08: Course Registration

The system shall allow students to register for applicable courses.

FR-09: Course Information

The system shall allow students to view their registered courses.

12.5 Attendance Management

FR-10: Attendance Entry

The system shall allow faculty members to record student attendance.

FR-11: Attendance Update

Authorized faculty members shall be able to update attendance records when corrections are required.

FR-12: Attendance Viewing

Students shall be able to view their attendance for individual courses.

FR-13: Attendance Summary

The system shall calculate and display the student's attendance percentage.

12.6 Examination Management

FR-14: Examination Management

Authorized examination staff shall be able to create and manage examination information.

FR-15: Examination Schedule

The system shall allow authorized users to manage examination schedules.

FR-16: Marks Entry

Authorized faculty or examination staff shall be able to enter student marks.

FR-17: Marks Update

Authorized users shall be able to correct marks when appropriate.

12.7 Result Management

FR-18: Result Generation

The system shall generate academic results using the recorded marks.

FR-19: Result Viewing

Students shall be able to view their academic results.

FR-20: Academic Performance

The system shall display relevant academic performance information to authorized users.

FR-21: Result Report

The system shall provide an academic result report for authorized users.

12.8 Report Management

FR-22: Academic Reports

The system shall allow authorized administrators to generate academic reports.

FR-23: Student Reports

The system shall provide student-specific academic information in report form.

FR-24: Report Access

Only authorized users shall be able to access reports containing academic information.

12.9 Notification Management

FR-25: Academic Notifications

The system should provide notifications for important academic events such as examinations, results and other relevant updates.

12.10 Administrative Functions

FR-26: User Management

Authorized administrators shall be able to manage system users and their roles.

FR-27: Academic Record Management

Authorized administrators shall be able to manage academic records.

FR-28: Dashboard

The system shall provide role-specific dashboards containing relevant information and functions.

13. Non-Functional Requirements Specification (NFRS)

13.1 Introduction

Non-functional requirements define the quality attributes and operational characteristics that the Student Record and Academic Management Platform should provide.

The major requirements considered are security, usability, reliability and scalability.

13.2 Security

NFR-01: Authentication Security

The system shall require authentication before allowing access to protected academic information.

NFR-02: Role-Based Authorization

Users shall only access functions and information permitted by their assigned roles.

NFR-03: Data Protection

Student academic information shall be protected against unauthorized access.

NFR-04: Secure Data Transmission

Sensitive information should be transmitted using secure communication mechanisms.

13.3 Usability

NFR-05: Simple Interface

The platform should provide a simple and understandable user interface.

NFR-06: Easy Navigation

Users should be able to reach frequently used features with minimal navigation.

NFR-07: Role-Based Interface

Users should see features relevant to their responsibilities.

NFR-08: Accessibility

The platform should be designed so that users can easily understand and interact with the system.

13.4 Reliability

NFR-09: Accurate Records

The system should maintain accurate and consistent academic records.

NFR-10: Data Integrity

The system should prevent invalid or inconsistent academic information from being stored.

NFR-11: Backup

Important academic data should be backed up to reduce the risk of data loss.

NFR-12: Error Handling

The system should provide appropriate error messages when operations fail.

13.5 Performance

NFR-13: Response Time

Common operations such as searching for student records should respond within an acceptable time.

NFR-14: Concurrent Users

The system should support multiple users accessing the platform simultaneously.

13.6 Scalability

NFR-15: User Scalability

The system should be capable of supporting an increasing number of students, faculty members and administrators.

NFR-16: Data Scalability

The database should be capable of handling increasing academic records over multiple academic years.

13.7 Maintainability

NFR-17: Modular Design

The system should use a modular architecture so that individual components can be maintained and improved independently.

NFR-18: Documentation

Important system components and development processes should be properly documented.

13.8 Availability

NFR-19: System Availability

The platform should be available to authorized users whenever academic services are required, subject to planned maintenance.

13.9 Privacy

NFR-20: Academic Data Privacy

Personal and academic information should only be accessible to authorized users according to institutional policies.

14. Product Vision Document

14.1 Product Name

Enterprise Student Record and Academic Management Platform

14.2 Product Vision Statement

To provide a centralized, secure and user-friendly platform that simplifies student record and academic information management for higher education institutions, while improving accessibility, accuracy and efficiency for students, faculty and administrators.

14.3 Product Mission

The mission of the platform is to:

- **Centralize student academic information**
- **Simplify academic record management**
- **Provide students with easy access to their academic information**
- **Help faculty efficiently manage attendance and marks**
- **Support administrators in managing academic records**
- **Improve examination and result management**
- **Reduce manual work and data-entry errors**
- **Provide reliable academic reporting**
- **Maintain appropriate security and access control**

14.4 Product Objectives

The major objectives of the product are:

1. **Develop a centralized student information system.**
2. **Simplify student registration and course registration.**
3. **Digitize attendance management.**
4. **Support examination and marks management.**
5. **Automate academic result generation.**
6. **Provide academic reports.**
7. **Implement role-based access.**
8. **Improve accessibility of academic information.**
9. **Reduce repetitive administrative work.**
10. **Provide a scalable foundation for future development.**

14.5 Target Users

The primary target users are:

- **Students**
- **Faculty members**
- **Academic administrators**
- **Examination cell staff**
- **IT support personnel**
- **College management**

14.6 Product Value Proposition

The platform provides a single centralized environment for managing academic information instead of relying on disconnected or manual processes.

For Students

Easy access to:

- Profile
- Courses
- Attendance
- Examinations
- Results
- Academic reports

For Faculty

Efficient tools for:

- Student management
- Attendance
- Marks entry
- Academic monitoring
- Reports

For Administrators

Centralized tools for:

- Student records
- Course management
- Academic records
- Reporting
- User management

For Examination Cell

Tools for:

- Examination management
- Marks management
- Result generation
- Result reporting

14.7 Business Objectives

The product should help the institution:

- Reduce manual academic record management
- Improve data accuracy
- Reduce administrative workload
- Improve access to academic information
- Improve academic reporting
- Support better academic decision-making
- Provide a foundation for future digital services

14.8 Success Criteria

The product will be considered successful when:

- **Students can easily access their academic information.**
- **Faculty can record attendance efficiently.**
- **Authorized users can manage marks and results.**
- **Administrators can manage student records centrally.**
- **Academic reports can be generated efficiently.**
- **Users can access only information appropriate to their roles.**
- **The system maintains reliable and consistent academic information.**
- **Users provide positive feedback about usability.**

15. Minimum Viable Product (MVP)

15.1 MVP Definition

A Minimum Viable Product is the first usable version of a product containing the essential features required to solve the primary user problem.

For this project, the MVP focuses on the core academic management functions rather than advanced features.

15.2 MVP Objective

The objective of the MVP is to provide a working foundation for centralized management of student academic information.

15.3 MVP Features

1. User Login

Users can securely log in according to their assigned role.

2. Student Registration

Authorized administrators can register students and maintain their basic academic profiles.

3. Student Record Management

The system stores and manages student academic information.

4. Course Registration

Students can view and register for applicable courses.

5. Attendance Management

Faculty can record attendance and students can view their attendance.

6. Examination Management

Authorized users can manage examination information.

7. Marks Management

Faculty or authorized examination staff can enter and manage marks.

8. Result Generation

The system can generate academic results using recorded marks.

9. Academic Reports

Authorized users can generate and view academic reports.

10. Role-Based Dashboards

Different users receive dashboards appropriate to their responsibilities.

15.4 MVP Feature Priority

Feature	Priority	MVP
User Login	High	Yes
Student Registration	High	Yes
Student Records	High	Yes
Course Registration	High	Yes
Attendance Management	High	Yes
Examination Management	High	Yes
Marks Management	High	Yes
Result Generation	High	Yes
Academic Reports	High	Yes
Notifications	Medium	Optional
Advanced Analytics	Low	No

AI-Based Recommendations	Low	No
Online Fee Payment	Low	No

15.5 MVP Users

The first release will primarily support:

- **Students**
- **Faculty**
- **Academic Administrators**
- **Examination Cell Staff**

15.6 Future Features

Features that can be considered after the MVP include:

- **Mobile application**
- **Advanced analytics**
- **AI-based academic assistance**
- **Predictive student performance analysis**
- **Online fee payment**
- **Parent/guardian access**
- **Advanced notification system**
- **Integration with other institutional systems**

15.7 MVP Success Criteria

The MVP will be successful if users can:

1. **Log into the platform.**
2. **Register and manage student records.**
3. **Register and view courses.**
4. **Record and view attendance.**
5. **Manage examinations and marks.**
6. **Generate and view results.**
7. **Generate basic academic reports.**
8. **Access features according to their assigned role.**

16. Product Roadmap

16.1 Roadmap Overview

The product roadmap describes the planned development of the Student Record and Academic Management Platform in different phases. The roadmap begins with understanding the problem and users and gradually progresses toward design, development, testing and deployment.

16.2 Phase 1 — Domain Analysis and Problem Understanding

Activities

- Understand the academic management process
- Identify stakeholders
- Study user needs and pain points
- Define the business problem
- Prepare the Business Requirements Document
- Define the project scope

Expected Outcome

A clear understanding of the academic domain, users and business problem.

16.3 Phase 2 — Design Thinking and Prototyping

Activities

- Empathize with users
- Define the core problem
- Generate solution ideas
- Create wireframes
- Develop the initial prototype
- Test the prototype
- Collect user feedback

Expected Outcome

A validated product concept and initial user interface design.

16.4 Phase 3 — Agile Sprint Planning

Activities

- Convert requirements into user stories
- Prioritize the product backlog
- Define MVP tasks
- Plan development sprints
- Assign responsibilities
- Define sprint goals

Expected Outcome

A prioritized Agile product backlog and development plan.

16.5 Phase 4 — Core Application Development

Activities

- **Design the database**
- **Develop backend services**
- **Develop frontend interfaces**
- **Implement authentication**
- **Implement student management**
- **Implement course management**
- **Implement attendance**
- **Implement examination and result modules**

Expected Outcome

A functional MVP of the Student Record and Academic Management Platform.

16.6 Phase 5 — Testing and Quality Assurance

Activities

- **Unit testing**
- **Integration testing**
- **System testing**
- **User acceptance testing**
- **Security testing**
- **Performance testing**
- **Bug flxing**

Expected Outcome

A stable and tested application suitable for deployment.

16.7 Phase 6 — DevOps and Deployment

Activities

- **Git repository management**
- **Continuous Integration**
- **Continuous Deployment**
- **Containerization**
- **Application deployment**
- **Monitoring and logging**

Expected Outcome

An automated and maintainable deployment process.

16.8 Phase 7 — Continuous Improvement

Activities

- **Monitor system usage**
- **Collect user feedback**
- **Analyze system performance**
- **Identify improvements**
- **Release new features**
- **Improve usability and reliability**

Expected Outcome

Continuous improvement of the product based on user needs and system performance.

16.9 Roadmap Summary

Phase	Main Focus	Major Deliverable
Phase 1	Domain Analysis	Business understanding and requirements
Phase 2	Design Thinking	Prototype
Phase 3	Agile Planning	Product backlog and sprint plan
Phase 4	Development	Functional MVP
Phase 5	Testing	Tested application
Phase 6	DevOps	Deployed application
Phase 7	Improvement	Continuous product enhancement

17. Project Charter

17.1 Project Name

17.2 Project Purpose

The purpose of this project is to design a centralized software platform for managing student records and academic information within a higher education institution.

17.3 Project Problem

Academic information may be managed through separate or manual processes, which can result in delays, data-entry errors, difficulty accessing information and increased administrative workload.

17.4 Project Objective

The project aims to design a secure, centralized and user-friendly platform for managing:

- Student records
- Student registration
- Course registration
- Attendance
- Examinations
- Marks
- Results
- Academic reports

17.5 Project Scope

In Scope

- User authentication
- Student management
- Course management
- Attendance management
- Examination management
- Marks management
- Result generation
- Academic reporting
- Role-based access

Out of Scope for Initial Release

- Online fee payment
- Hostel management
- Transportation management
- Library management
- Advanced AI features

- Advanced predictive analytics

17.6 Major Stakeholders

- Students
- Faculty members
- Academic administrators
- Examination cell staff
- IT support personnel
- College management

17.7 Major Deliverables

The project will produce:

1. Business Requirements Document
2. Product Vision Document
3. Stakeholder Analysis
4. User Persona Report
5. Functional Requirements Specification
6. Non-Functional Requirements Specification
7. Design Thinking Report
8. IBM Design Thinking Report
9. IBM Loop Documentation
10. MVP Definition
11. Product Roadmap
12. System Architecture
13. Project Charter
14. Product Discovery Documentation
15. Initialized Git Repository
16. Standardized Project Folder Structure

17.8 Project Constraints

The major constraints include:

- Limited development time
- Availability of accurate academic data
- Institutional policies
- Data privacy requirements
- Infrastructure limitations
- User adoption
- Technical resources

17.9 Project Assumptions

The project assumes that:

- Users have access to the required computing devices.
- Users have valid credentials.
- Academic departments provide accurate information.
- Authorized staff maintain academic records.
- Suitable network infrastructure is available.
- Users can provide feedback during testing.

17.10 Project Success Criteria

The project will be considered successful when:

- The major user needs are clearly identified.
- Requirements are properly documented.
- A clear product vision is established.
- The MVP scope is defined.
- The initial prototype is understandable and usable.
- The system architecture is clearly documented.
- The project is prepared for Agile development.
- The Git repository and project structure are initialized.

17.11 Future Development

After the MVP, the platform can be extended with additional services such as mobile applications, advanced analytics, AI-based academic assistance, predictive analysis and integration with other institutional systems.

18. Initial System Architecture

18.1 Architecture Overview

The Student Record and Academic Management Platform will use a layered architecture consisting of a user interface layer, application layer and database layer.

The architecture is designed to separate user interaction, business logic and data storage. This makes the system easier to maintain, test and extend in future development phases.

18.2 Architecture Components

1. User Interface Layer

The User Interface Layer provides access to the platform for different users.

Users include:

- Students
- Faculty members
- Academic administrators

- Examination cell staff
- IT support personnel

The interface will provide role-specific dashboards and functions.

2. Application Layer

The Application Layer contains the business logic of the platform.

Major modules include:

- Authentication Module
- Student Management Module
- Course Management Module
- Attendance Management Module
- Examination Module
- Marks Management Module
- Result Management Module
- Report Generation Module
- Notification Module

3. Database Layer

The Database Layer stores and manages academic information.

Major data entities include:

- Users
- Students
- Faculty
- Courses
- Enrollments
- Attendance
- Examinations
- Marks
- Results
- Academic Reports

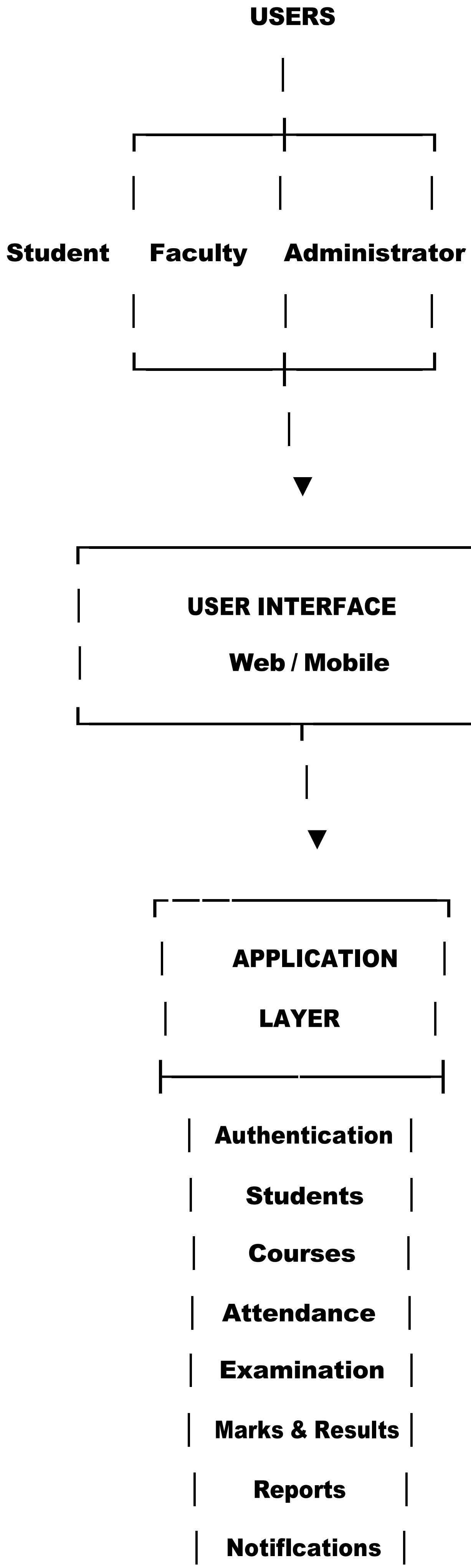
18.3 Architecture Flow

The basic system flow is:

User → User Interface → Application Layer → Database

The application processes user requests, applies business rules and retrieves or updates the required information in the database.

18.4 Initial Architecture Diagram


USERS
Student
Faculty
Administrator
USER INTERFACE
Web / Mobile
APPLICATION
LAYER
Authentication
Students
Courses
Attendance
Examination
Marks & Results
Reports
Notifications

```
└──────────┬────── ──┘
           │
           ▼
┌──────────────────────────────────────┐
│              DATABASE                │
├──────────────────────────────────────┤
          │   Users        │
          │  Students      │
          │   Faculty      │
          │   Courses      │
          │  Attendance    │
          │ Examinations   │
          │    Marks       │
          │   Results      │
          └────────────────┘
```

18.5 Security Considerations

The architecture will include role-based access so that users can access only the information and functions appropriate to their responsibilities.

Sensitive academic information should be protected through authentication, authorization and secure communication.

18.6 Future Architecture Expansion

In later development phases, the architecture can be expanded with:

- API gateway
- Cloud deployment
- Containerization
- CI/CD pipeline
- Monitoring
- Logging
- Automated testing
- Infrastructure automation

These components can be introduced as the project progresses through later engineering and DevOps phases.

19. Project Repository and Folder Structure

19.1 Repository Name

student-academic-management-platform

19.2 Purpose

The repository will contain the source code, documentation, tests, configuration and deployment resources required for the Student Record and Academic Management Platform.

19.3 Proposed Folder Structure

```
student-academic-management-platform/
│
├── .github/
│   └── workflows/
│
├── app/
│   ├── frontend/
│   ├── backend/
│   ├── modules/
│   │   ├── students/
│   │   ├── courses/
│   │   ├── attendance/
│   │   ├── examination/
│   │   └── results/
│   │
│   ├── database/
│   └── tests/
```

```
│
├── docs/
│   ├── BRD/
│   ├── FRS/
│   ├── NFRS/
│   ├── personas/
│   ├──design-thinking/
│   ├──ibm-design-thinking/
│   ├── ibm-loop/
│   ├── architecture/
│   └── roadmap/
│
├── docker/
│
├── deployment/
│
├── monitoring/
│
├── README.md
│
└── .gitignore
```

19.4 Folder Description

Folder	Purpose
.github/	GitHub Actions and CI/CD workflows
app/frontend/	Frontend application

app/backend/	Backend application
app/modules/	Application business modules
app/database/	Database schemas and related files
app/tests/	Automated tests
docs/	Project documentation
docker/	Docker configuration
deployment/	Deployment configuration
monitoring/	Monitoring-related configuration
README.md	Project overview and setup instructions
.gitignore	Files excluded from Git

20. Git Repository Initialization

The project repository will be initialized using Git to maintain version history and support collaborative development.

Basic Git Initialization

mkdir student-academic-management-platform

cd student-academic-management-platform

git init

git add .

git commit -m "Initial project structure"

The repository can then be connected to a remote GitHub repository.

Initial Commit

The first commit will contain:

- Project documentation
- Folder structure
- README file
- Initial architecture documentation
- Requirements documentation
- Project planning documents

Purpose of Version Control

Git will help the project team to:

- Track changes
- Maintain project history
- Collaborate between team members
- Create development branches
- Review changes
- Recover previous versions
- Support future CI/CD implementation

21. Product Discovery Documentation

21.1 Purpose of Product Discovery

Product Discovery is the process of understanding the users, their problems, business needs and possible solutions before beginning full-scale product development.

For the Student Record and Academic Management Platform, Product Discovery helps ensure that the proposed solution addresses real academic management problems.

21.2 Discovery Objectives

The main objectives of Product Discovery are:

- Understand the academic management process.
- Identify the major users of the system.
- Understand user problems and expectations.
- Identify important business requirements.
- Determine the most valuable product features.
- Validate the proposed solution.
- Define the initial MVP.
- Reduce the risk of developing unnecessary features.

21.3 Key Users Identified

The discovery process identified the following major users:

Students

Students require quick access to their academic information such as courses, attendance, examinations and results.

Faculty Members

Faculty members require efficient tools for attendance, marks and student academic management.

Academic Administrators

Administrators require centralized student records, course information and academic reports.

Examination Cell

The examination cell requires tools for examination scheduling, marks management and result generation.

IT Support

IT support personnel require a secure, reliable and maintainable system.

21.4 User Problems

User	Major Problem
Student	Difficulty accessing academic information
Student	Difficulty monitoring attendance
Student	Dependence on different sources for academic information
Faculty	Time-consuming attendance management
Faculty	Manual marks entry and report preparation
Administrator	Managing large volumes of student records

Administrator	Duplicate or inconsistent information
Examination Cell	Managing examination and result information
IT Support	Maintaining system security and reliability

21.5 User Needs

Based on the identified problems, the following user needs were identified:

1. Easy access to academic information.
2. Centralized student records.
3. Simple attendance management.
4. Efficient course registration.
5. Easy examination information access.
6. Reliable marks and result management.
7. Efficient academic reporting.
8. Secure access to academic information.
9. Simple and understandable user interfaces.
10. Role-specific functionality.

21.6 Opportunity Areas

The identified problems provide opportunities to improve the academic management process.

Opportunity 1 — Centralized Information

Create a single platform where academic information can be accessed and managed.

Opportunity 2 — Self-Service for Students

Allow students to independently access attendance, courses and results.

Opportunity 3 — Digital Faculty Workflow

Provide faculty members with digital tools for attendance and marks management.

Opportunity 4 — Automated Reporting

Reduce manual effort by generating academic reports automatically.

Opportunity 5 — Role-Based Access

Provide users with only the functions and information relevant to their roles.

21.7 How Might We Questions

The following How Might We questions were created during Product Discovery:

HMW 1

How might we make academic information easily accessible to students?

HMW 2

How might we reduce the time faculty members spend managing attendance and marks?

HMW 3

How might we help administrators maintain accurate student records?

HMW 4

How might we simplify examination and result management?

HMW 5

How might we provide secure access to academic information for different users?

21.8 Solution Ideas

Based on the identified opportunities, the following solutions were considered:

Problem	Solution
Difficult information access	Centralized dashboard
Attendance monitoring	Digital attendance module
Manual course registration	Online course registration
Manual marks management	Digital marks entry
Result preparation	Automated result generation
Difficult report preparation	Automated reporting
Different user needs	Role-based dashboards

Security concerns	Authentication and authorization

21.9 Feature Prioritization

Features were prioritized according to their importance to users and their contribution to solving the core problem.

Must Have

- Login and authentication
- Student registration
- Student records
- Course registration
- Attendance management
- Examination management
- Marks management
- Result generation
- Academic reports
- Role-based access

Should Have

- Notifications
- Search and filtering
- Dashboard analytics
- Profile management

Could Have

- Mobile application
- Advanced analytics
- AI-based academic assistance
- Predictive performance analysis

Won't Have in MVP

- Online fee payment
- Hostel management
- Transportation management
- Library management

21.10 Product Discovery Outcome

The Product Discovery process established that the primary requirement is a centralized and secure platform for managing student academic information.

The discovery process also helped prioritize the MVP around the most important academic workflows:

Student Records → Course Registration → Attendance → Examination → Results → Academic Reports

The findings provide the foundation for product design, Agile planning and future development.

21.11 Validation Plan

The proposed product concept can be validated through:

- **User interviews**
- **Surveys**
- **Prototype testing**
- **Usability testing**
- **Faculty feedback**
- **Administrator feedback**
- **Student feedback**

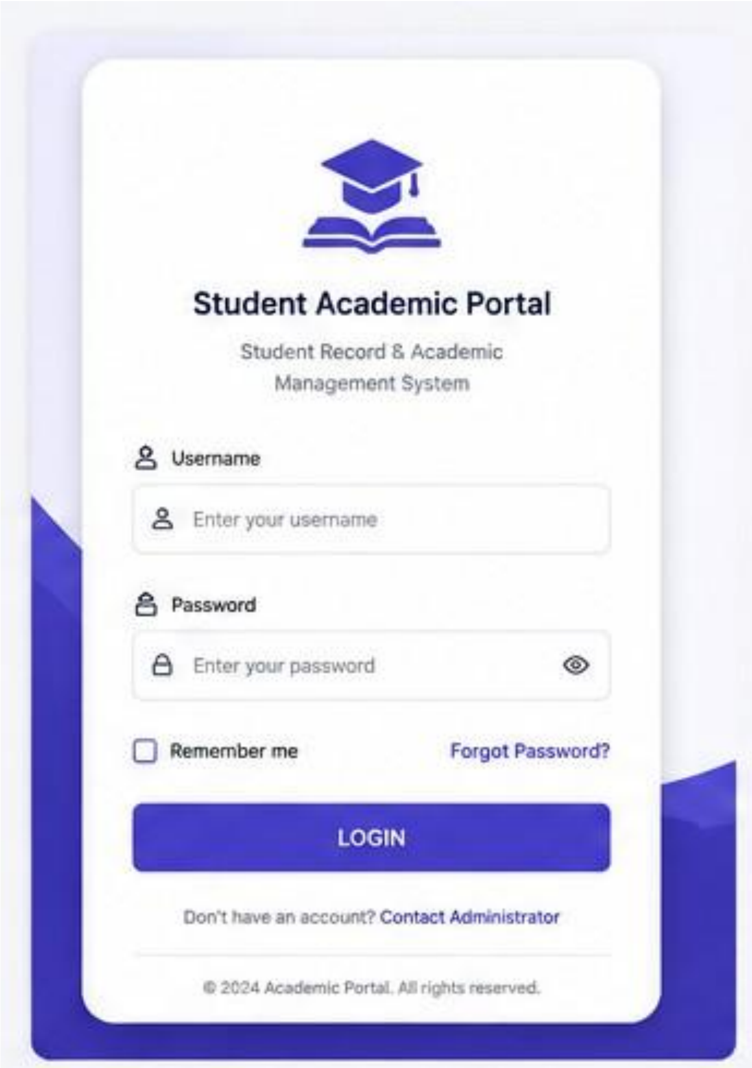
Feedback collected during validation will be used to improve the product before full-scale development.

22. Prototype / Wireframes

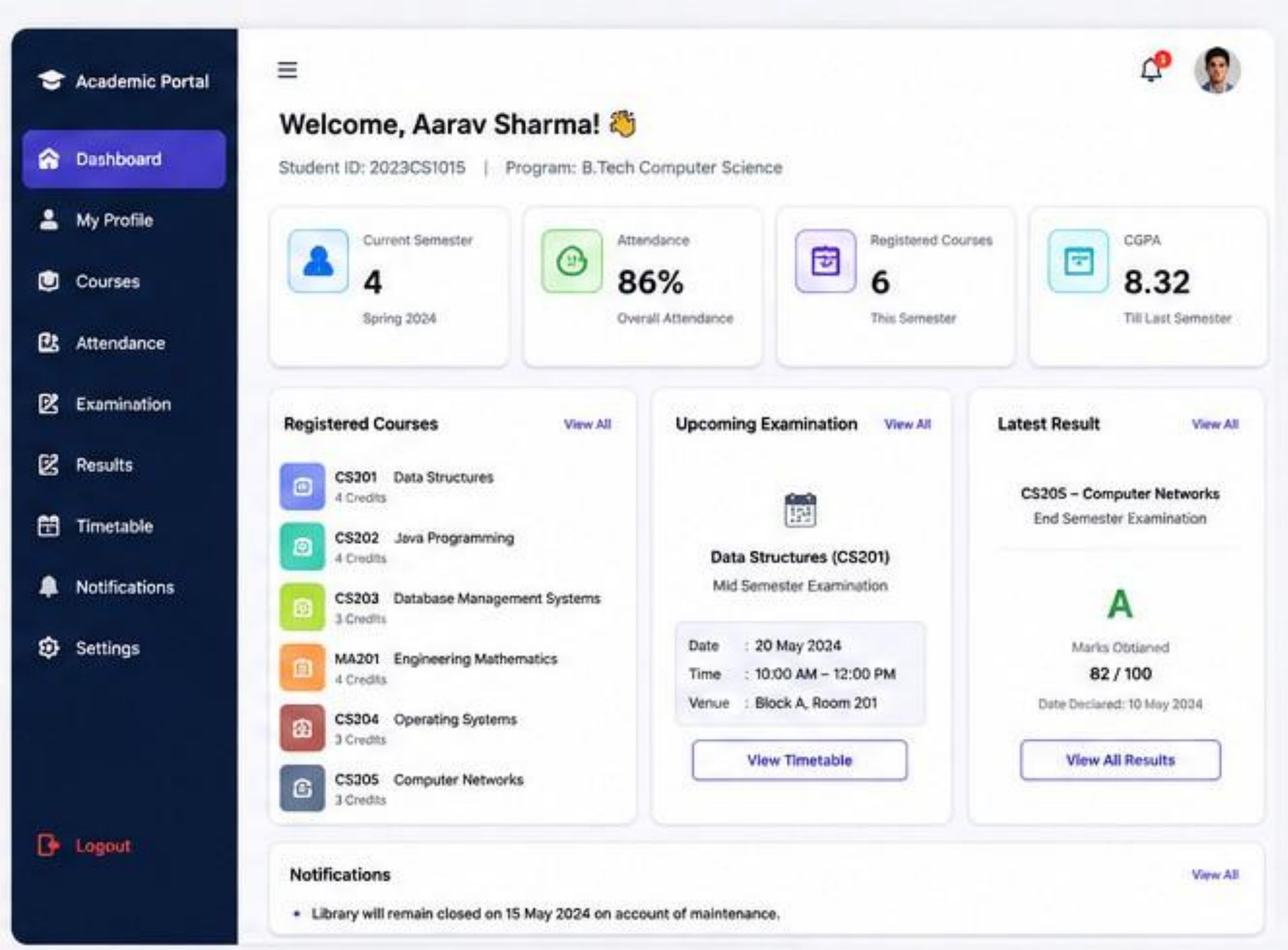
Then add this short introduction:

The following low-fidelity/high-fidelity prototype screens represent the proposed user interface of the Student Record and Academic Management Platform. The screens demonstrate the major workflows identified during Product Discovery and MVP definition.

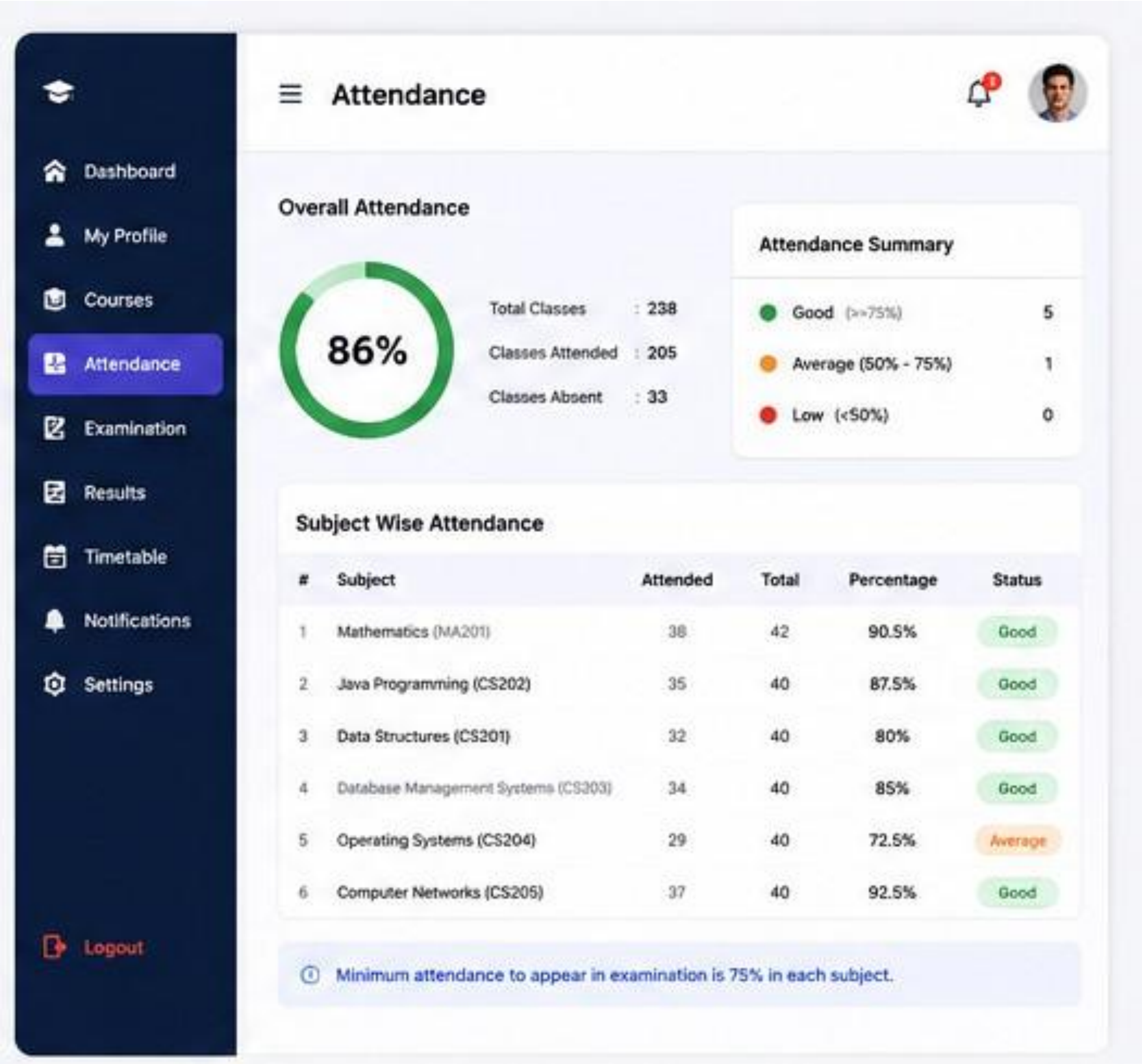
22.1 Login Screen



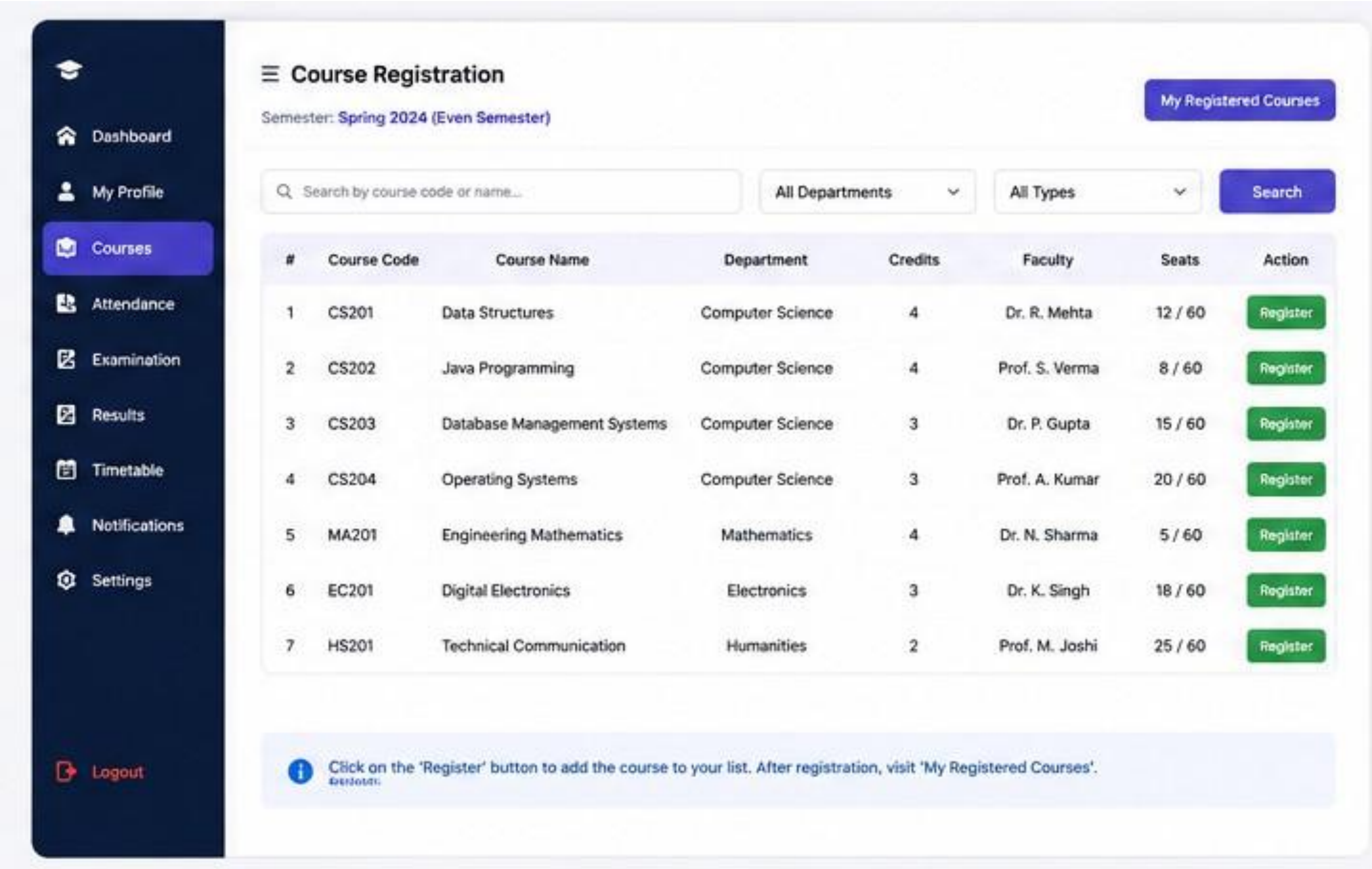
22.2 Student Dashboard



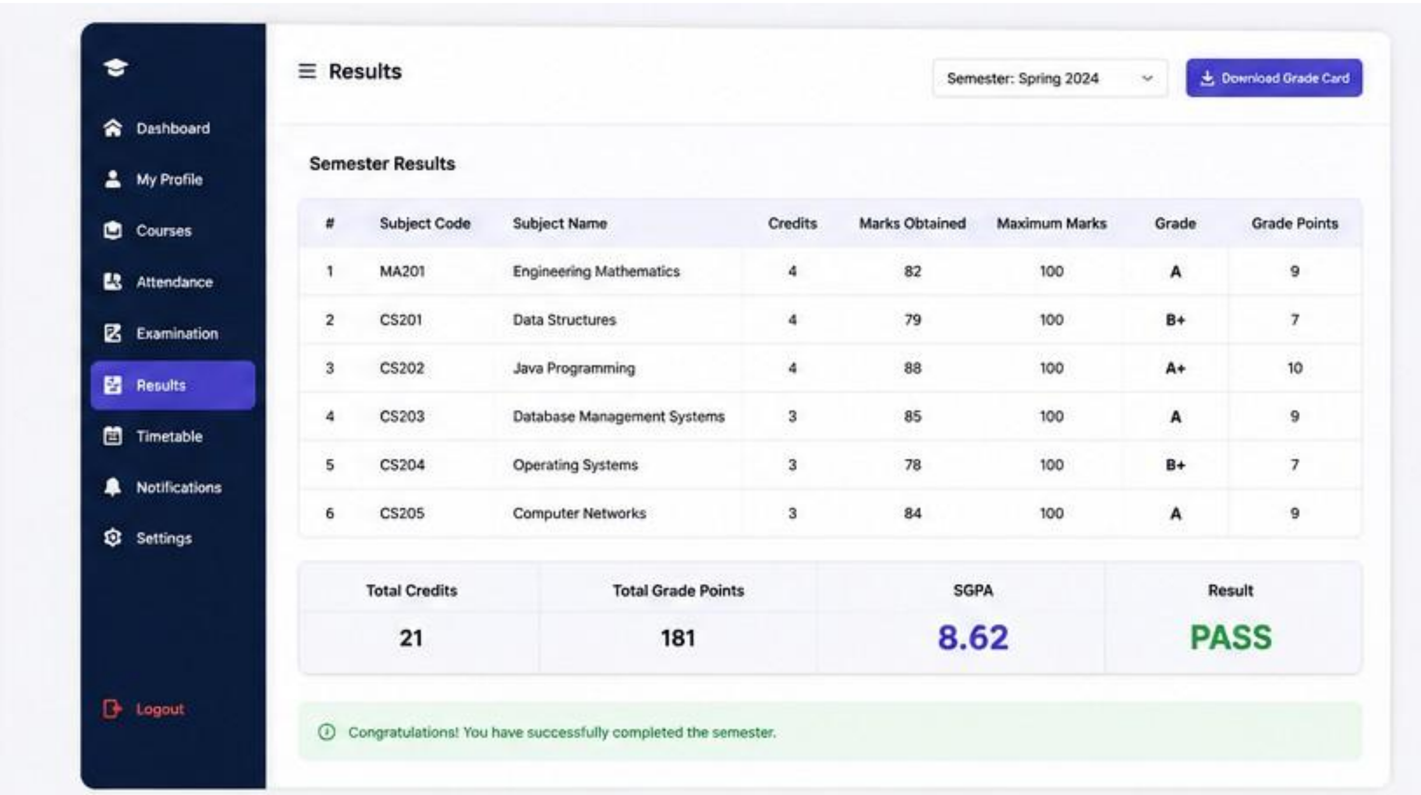
22.3 Attendance Screen



22.4 Course Registration Screen



22.5 Results Screen



23. System Architecture Diagram

USERS

STUDENT FACULTY ADMINISTRATOR

↓

USER INTERFACE
Web Portal

↓

APPLICATION
LAYER

Authentication
Student Module
Course Module
Attendance
Examination
Marks & Results
Reports

↓

```
┌──────────────────────────────┐
        │  DATABASE  │
├──────────────────────────────┤
        │ Students    │
        │ Faculty     │
        │ Courses     │
        │ Attendance  │
        │ Examinations │
        │ Marks       │
        │ Results     │
└──────────────────────────────┘
```

24. Product Development Roadmap

24.1 Roadmap Overview

The product roadmap describes the planned development of the Enterprise Student Record and Academic Management Platform from initial problem understanding to continuous improvement.

24.2 Development Phases

Phase 1 — Domain Analysis

- Understand the academic management domain
- Identify stakeholders
- Identify user problems
- Define the business problem

Deliverable: Business understanding and requirements

Phase 2 — Design Thinking

- Empathize with users
- Define user problems
- Generate possible solutions
- Develop prototypes
- Validate the proposed solution

Deliverable: User-centered product concept and prototype

Phase 3 — Agile Planning

- **Create product backlog**
- **Deflne user stories**
- **Prioritize requirements**
- **Plan development sprints**

Deliverable: Agile development plan

Phase 4 — MVP Development

- **Develop authentication**
- **Develop student management**
- **Develop course management**
- **Develop attendance management**
- **Develop examination and result modules**
- **Develop academic reporting**

Deliverable: Functional MVP

Phase 5 — Testing and Quality Assurance

- **Unit testing**
- **Integration testing**
- **System testing**
- **Usability testing**
- **Security testing**
- **Bug flxing**

Deliverable: Tested and stable application

Phase 6 — Deployment

- **Conflgure deployment environment**
- **Set up CI/CD**
- **Deploy the application**
- **Conflgure monitoring and logging**

Deliverable: Deployed application

Phase 7 — Continuous Improvement

- **Collect user feedback**
- **Monitor system performance**
- **Identify improvements**
- **Release new features**
- **Improve usability and reliability**

Deliverable: Continuously improved product

24.3 Roadmap Summary

Phase	Focus	Main Deliverable
1	Domain Analysis	Requirements
2	Design Thinking	Prototype
3	Agile Planning	Product Backlog
4	MVP Development	Working MVP
5	Testing	Tested Application
6	Deployment	Deployed Product
7	Improvement	Product Enhancements

25. Project Repository Structure

25.1 Repository Name

student-academic-management-platform

25.2 Repository Purpose

The repository will contain the documentation, source code, database resources, testing files and deployment configuration required for the Student Record and Academic Management Platform.

25.3 Folder Structure

```
student-academic-management-platform/
│
├── app/
│   ├── frontend/
```

```
│   ├── backend/
│   ├── database/
│   └── tests/
│
├── docs/
│   ├── BRD/
│   ├── FRS/
│   ├── NFRS/
│   ├── personas/
│   ├──design-thinking/
│   ├── architecture/
│   └── roadmap/
│
├── deployment/
├── docker/
├── monitoring/
│
├── README.md
└── .gitignore
```

25.4 Folder Description

Folder/File	Purpose
app/frontend	Frontend application
app/backend	Backend application
app/database	Database-related files
app/tests	Testing files

docs	Project documentation
deployment	Deployment conflguration
docker	Container conflguration
monitoring	Monitoring conflguration
README.md	Project overview and setup instructions
.gitignore	Files excluded from Git

26. Git Repository Initialization

Git will be used for version control and collaborative development of the project.

26.1 Git Initialization Commands

mkdir student-academic-management-platform

cd student-academic-management-platform

git init

git add .

git commit -m "Initial project structure"

26.2 Purpose of Git

Git will help the development team to:

- Track changes
- Maintain project history
- Collaborate between team members
- Create development branches
- Review changes
- Recover previous versions
- Support future CI/CD implementation

26.3 Initial Repository Contents

The initial repository will contain:

- Project documentation
- Requirements documentation
- Product vision
- User personas
- Design Thinking documentation
- Architecture documentation
- Project roadmap
- Initial project folder structure
- README file

27. Final Product Summary

The Enterprise Student Record and Academic Management Platform is a centralized academic management solution designed to simplify the management and accessibility of student academic information.

The platform provides different functionalities according to user roles.

Student

Students can:

- View their profile
- Register for courses
- Check attendance
- View examination information
- View marks and results
- Access academic reports

Faculty

Faculty members can:

- View student information
- Manage attendance
- Enter marks
- Monitor academic performance
- Access academic reports

Administrator

Administrators can:

- Register students
- Manage student records
- Manage courses
- Manage users
- Generate academic reports

Examination Cell

The examination cell can:

- Manage examinations
- Manage marks
- Generate results
- Access examination reports

28. Expected Benefits

The proposed platform is expected to provide the following benefits:

1. **Centralized Information**
   Academic information can be maintained in one platform.
2. **Reduced Manual Work**
   Digital workflows can reduce repetitive administrative activities.
3. **Improved Accuracy**
   Centralized records can reduce duplicate and inconsistent information.
4. **Faster Access**
   Students and staff can access required information quickly.
5. **Better Academic Monitoring**
   Attendance, marks and results can be monitored more efficiently.
6. **Improved Reporting**
   Academic reports can be generated from centralized records.
7. **Better Security**
   Role-based access can help protect academic information.
8. **Improved User Experience**
   Users can access features relevant to their specific roles.

29. Future Scope

The platform can be expanded in future development phases with additional features.

Possible future enhancements include:

- Mobile application
- Online fee payment
- Parent or guardian portal
- Library management
- Hostel management
- Transportation management
- Advanced academic analytics
- AI-based academic assistance
- Predictive student performance analysis

- Automated notifications
- Integration with existing university systems
- Cloud-based deployment

These features are intentionally excluded from the initial MVP so that the development team can focus on the core academic management requirements.

30. Conclusion

The Enterprise Student Record and Academic Management Platform was designed to address the challenges associated with managing student academic information through fragmented or manual processes.

The project applied Design Thinking to understand users, identify their problems and develop a user-centered solution. IBM Design Thinking principles and the IBM Loop were used to support continuous improvement and focus on user outcomes.

The project requirements were documented through the Business Requirements Document, Functional Requirements Specification and Non-Functional Requirements Specification. Stakeholder analysis, user personas, product vision, MVP definition, product roadmap and project charter were also prepared.

The proposed MVP focuses on the most important academic workflows, including student registration, student record management, course registration, attendance, examinations, marks, results and academic reporting.

A prototype was developed to demonstrate the major user interfaces, while the initial system architecture and repository structure provide a foundation for future development.

Overall, the project provides a structured foundation for developing a centralized, secure, scalable and user-friendly academic management platform using Agile, Product Engineering and DevOps practices.

31. Final Deliverables Checklist

Deliverable	Status
Introduction	✓
Business Understanding	✓
Problem Statement	✓
Stakeholder Analysis	✓

Stakeholder Matrix	✓
User Personas	✓
Design Thinking	✓
IBM Design Thinking	✓
IBM Loop	✓
BRD	✓
FRS	✓
NFRS	✓
Product Vision	✓
MVP	✓
Product Roadmap	✓
Project Charter	✓
System Architecture	✓
Repository Structure	✓
Product Discovery	✓
Validation Plan	✓
Prototype / Wireframes	✓
Git Initialization	✓
Future Scope	✓
Conclusion	✓